

ChemNovus — Product Spec Sheet (fictional)

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NovaTherm PA66-GF30

Product family: NovaTherm engineering polymers **Grade:** PA66-GF30 (30% glass-fiber reinforced polyamide 66) **Form:** Natural / black pellets, 25 kg bags or 1000 kg octabins **Product code:** NT-6630-N (natural), NT-6630-BK (black)

Description

High-stiffness, high-temperature engineering thermoplastic. 30% short-glass-fiber reinforcement for dimensional stability under mechanical and thermal load. Injection-molding grade for demanding under-hood and structural automotive parts.

Key properties (typical, dry-as-molded)

Property	Value	Method
Density	1.36 g/cm³	ISO 1183
Tensile strength	185 MPa	ISO 527
Tensile modulus	10,500 MPa	ISO 527
Elongation at break	3.0 %	ISO 527
Flexural modulus	9,800 MPa	ISO 178
Charpy notched impact (23 °C)	12 kJ/m²	ISO 179
HDT (1.8 MPa)	250 °C	ISO 75
Melting point	262 °C	ISO 11357
Continuous use temp (RTI)	150 °C	UL 746B

Property	Value	Method
Water absorption (23 °C, sat.)	5.5 %	ISO 62
Flammability	HB (UL 94)	—
Mold shrinkage (flow / cross)	0.4 % / 1.0 %	ISO 294

Processing

- Melt temp: 280–300 °C · Mold temp: 80–100 °C
- Pre-dry: 80 °C, 4 h, moisture < 0.2%
- Recommended for injection molding; regrind up to 15%

Applications

Engine mounts, intake manifolds, coolant/thermostat housings, connectors, brackets, EV busbar supports, gears.

Compliance

REACH compliant · RoHS compliant · UL 94 HB listed (file E-fictional) · IMDS entry available · **Not** food-contact or medical grade.

Commercial Notes

Best-fit customers: automotive tier-1/tier-2 molders; EV powertrain & thermal-management parts; industrial motor housings. Volume sweet spot 20–500 t/yr.